

Practical - 7

Practical Title: PHP Form Processing with Server-Side Validation and CSV/JSON File Storage

Problem definition: Process submitted registration/contact form data using PHP. Validate and sanitize inputs on the server side and store records in CSV or JSON file format. Display success/error messages.

Theory:

1. Is the form submitted using POST?

Answer: Yes. The form uses the POST method to send data to the PHP file securely without showing the data in the URL.

2. Are inputs validated and sanitized server side?

Answer: Yes. PHP checks whether all fields are filled, validates the email and phone number, and sanitizes the input before storing it.

3. Is file writing handled safely?

Answer: Yes. The JSON file is read first, the new record is added, and then the updated data is converted to JSON and stored using `file_put_contents()`.

4. Are success and error responses displayed clearly?

Answer: Yes. The program displays a success message when registration is completed and clear error messages when the entered data is invalid or missing.

A. Implementation/ Output/Result (Keep only one option which is suitable to practical)

HTML tags

Tag	Purpose
<!DOCTYPE html>	Tells the browser that the document uses HTML5
<html>	Defines the complete HTML document
<head>	Contains information about the webpage
<title>	Sets the title shown on the browser tab
<body>	Contains the visible content of the webpage
<h2>	Displays a heading
<form>	Creates the registration form

Tag	Purpose
<label>	Gives a name/label to an input field
<input>	Takes input from the user
<select>	Creates a dropdown list
<option>	Adds options inside the dropdown
 	Adds a line break
<p>	Displays a paragraph or message
<hr>	Creates a horizontal line
<table>	Creates a table to display records
<tr>	Creates a table row
<th>	Creates a table heading
<td>	Creates a table data cell

Important attributes

Attribute	Purpose
method="POST"	Sends form data using POST
name="name"	Gives a name to the input so PHP can access it
type="text"	Creates a text input
type="email"	Creates an email input
type="submit"	Creates the Register button
value	Stores the value of an input/option

PHP functions and concepts

PHP function/concept	Purpose
\$_POST	Gets data submitted from the form
\$_SERVER	Gets information about the request
trim()	Removes extra spaces
empty()	Checks whether a field is empty
filter_var()	Validates the email
preg_match()	Checks the phone number format
htmlspecialchars()	Sanitizes/safely displays input
file_get_contents()	Reads data from the JSON file
json_decode()	Converts JSON into a PHP array
json_encode()	Converts PHP array into JSON
file_put_contents()	Writes data into the JSON file
array()	Creates an array
foreach()	Loops through stored student records
echo	Displays output on the webpage
if / elseif / else	Performs conditions

JSON

JSON concept

Purpose

[] Represents an empty JSON array

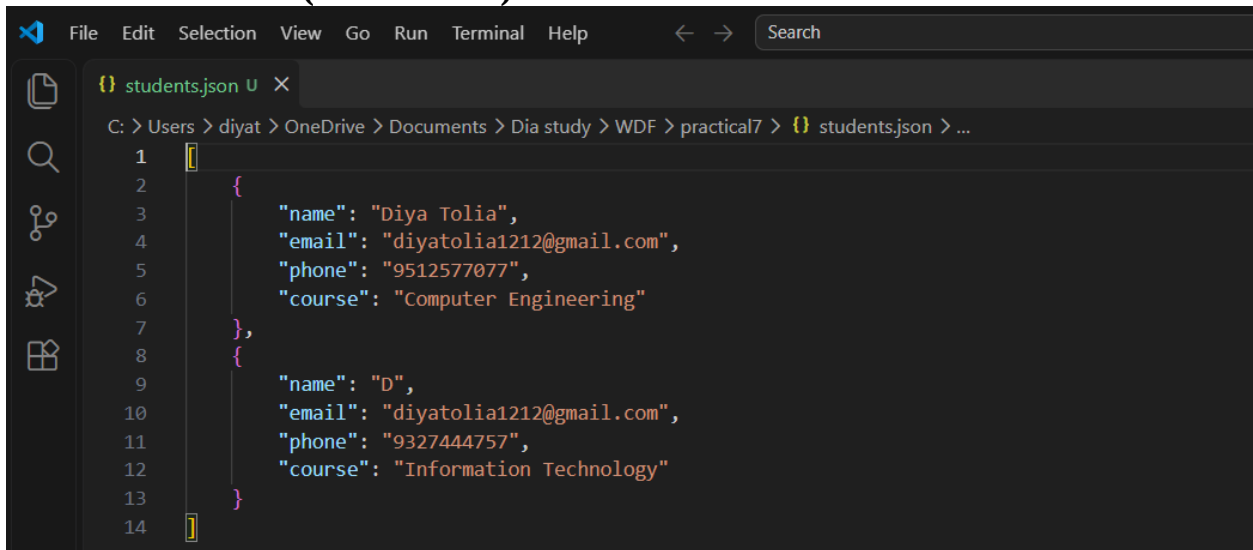
{ } Represents a JSON object

"name": "Diya" Stores data as key-value pair

json_decode() JSON → PHP array

json_encode() PHP array → JSON

A. Evidence of Work (Screenshots)



```

1  [
2      {
3          "name": "Diya Tolia",
4          "email": "diyatolia1212@gmail.com",
5          "phone": "9512577077",
6          "course": "Computer Engineering"
7      },
8      {
9          "name": "D",
10         "email": "diyatolia1212@gmail.com",
11         "phone": "9327444757",
12         "course": "Information Technology"
13     }
14 ]
  
```

First, it was blank and after we enter the data in form, the entered data will be visible in .json file



Student Registration Form

Registration successful!

Name:

Email:

Phone:

Course:

Stored Student Records

Name	Email	Phone	Course
Diya Tolia	diyatolia1212@gmail.com	9512577077	Computer Engineering

Student Registration Form

Registration successful!

Name:

Email:

Phone:

Course:

Stored Student Records

Name	Email	Phone	Course
Diya Tolia	diyatolia1212@gmail.com	9512577077	Computer Engineering
D	diyatolia1212@gmail.com	9327444757	Information Technology

A. GitHub Repository Link:

<https://github.com/diyatolia1212/StudentHub/tree/dab3b221b0724917b3ba8887248d2e9dcfa3fe33/practical7>

B. Challenges Faced:

- Understanding how to receive form data using the POST method.
- Validating the user input correctly in PHP.
- Understanding how to read and write data in a JSON file.
- Handling errors when fields are empty or invalid.
 - Converting PHP arrays into JSON format and storing them safely.

Learning Outcomes:

- Learned how to create and process forms using PHP.
- Learned how to use POST for sending form data.
- Learned how to validate and sanitize user inputs on the server side.
- Learned how to store student records in a JSON file.
- Learned how to use `json_encode()` and `json_decode()`.

- Learned how to display stored JSON records on a webpage.

Faculty Remarks: (Keep this blank for faculty)